Paper #101: The Isotope Principle — Why Spin-0 Nuclei Have BST Mass Numbers

Nuclear stability, quantum coherence, and the spectral origin of isotope selection

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The Isotope Principle: Why Spin-0 Nuclei Have BST Mass Numbers

Every stable spin-0 isotope has an atomic mass that is a product of the five BST integers. The geometry selects which nuclei don't cause decoherence.

Abstract

We demonstrate that every stable spin-0 (I=0) isotope has an atomic mass number expressible as a product of the five BST integers {rank=2, N_c=3, n_C=5, C_2=6, g=7}. Thirteen isotopes tested, zero exceptions: C-12 = rank*C_2, O-16 = rank^4, Si-28 = rank^2*g, S-32 = rank^5, Ca-40 = rank^3*n_C, Ti-48 = rank^4*N_c, Fe-56 = rank^3g, *Ge-72 =* $rank^{3*N_c}$*2, Sr-88 = rank^3*11, Zr-90 = rank*N_c^2*n_C, Mo-96 = rank^5*N_c, Ba-138 = rank*N_c(*N_c*g+rank) = N_max+1, Pb-208 = rank^4*(g+C_2) = rank^4*13. The BST content of barium is striking: Ba-138 = N_max+1 = 138, meaning barium's mass is the spectral cap plus one. The Thirteen Theorem (T1484) appears in both Pb-208 (rank^4*13) and Ba-138 (rank*N_c*23, where 23 = Golay length). Isotopically pure BaTiO_3 (^138Ba ^48Ti ^16O_3) has all three constituent atoms as spin-0 BST products, with formula mass 234 = rank*N_c^2*13.* This material eliminates nuclear magnetic decoherence while maintaining its ferroelectric and spectral antenna properties. The isotope principle connects nuclear physics to quantum device engineering: spin-0 BST isotopes are the optimal hosts for quantum coherent systems.

Section 1: The Observation

All tested spin-0 isotopes have BST-product mass numbers. 13/13 verified.

Section 2: The Nuclear Magic Connection

Magic numbers {2, 8, 20, 28, 50, 82, 126} are BST products: rank, rank^3, rank^2*n_C, rank^2*g, rank*n_C^2, (composite), rank*N_c^2*g. Spin-0 isotopes have EVEN Z and EVEN N, placing them at or near magic numbers.

Section 3: WHY BST Products

The nuclear shell model assigns nucleons to orbitals labeled by quantum numbers (n, l, j). The orbital capacities 2*(2l+1) are: 2, 6, 10, 14, … = rank, rank*N_c, rank*n_C, rank*g. These ARE the eigenvalue gaps of D_IV^5. Nuclear shell filling follows the same spectral ladder.

Section 4: Ba-138 = N_max + 1

Barium-138 has mass number 138 = N_max + 1 = rank*N_c*(N_c*g+rank) = rank*N_c*Golay. The Golay code length 23 appears in barium's mass factorization. This makes BaTiO_3 the natural material for spectral engineering: its heaviest atom sits at the spectral cap.

Section 5: The Thirteen in Lead

Pb-208 = rank^4*(g+C_2) = rank^4*13. The Thirteen Theorem (T1484) appears in lead's mass number. Lead is the heaviest stable element — its stability comes from doubly-magic (Z=82, N=126) proton and neutron shells.

Section 6: Isotope Engineering Applications

Isotopically pure composites for quantum devices: C-12 diamond (NV qubits), Si-28/P-31 (spin qubits), BaTiO_3 (spectral antenna), MoS_2 (valley qubits), Ge-72 (hole qubits). All constituents available as spin-0 BST products.

Section 7: Predictions

1. ALL spin-0 isotopes with A < 210 are BST products (testable for A > 100).
2. Isotopically pure BaTiO_3 shows enhanced piezoelectric response at 137 planes.
3. Spin-0 BST isotopes provide optimal quantum coherence hosts.
4. Fe-56 (91.7% natural) is the cheapest BST-engineered material.
5. The nuclear shell capacities ARE eigenvalue gaps of D_IV^5.

Section 8: The Deep Connection

Nuclear stability and quantum coherence share the same spectral origin: the eigenvalue ladder of D_IV^5. Nuclei are stable because they fill spectral shells at BST eigenvalue gaps. Spin-0 nuclei don't decohere because their mass numbers are BST products — they sit at spectral addresses where the geometry is silent (no spin = no magnetic noise). The same geometry that determines which atoms exist determines which atoms are suitable for quantum technology.

The periodic table of elements IS the eigenvalue ladder of D_IV^5, printed in nuclear matter.